

Generating Co-Speech Gestures for Virtual Agents from Multimodal Information Based on Transformer

Yue Yu*

Beijing Institute of Technology, China

Jiande Shi†

Beijing Institute of Technology, China

ABSTRACT

To generate co-speech gestures for virtual agents and enhance the correlation between gestures and input modalities, we propose a Transformer-based model, which encodes four-modal-like information (Audio Waveform, Mel-Spectrogram, Text, and SpeakerIDs). For the Mel-Spectrogram modal, we design a Mel-Spectrogram encoder based on the Swin Transformer pre-trained model to extract the audio spectrum features hierarchically. For the Text modal, we use the Transformer encoder to extract text features aligned with the audio. We evaluate on the TED-Gesture dataset. Compared with the state-of-art methods, we improve the mean absolute joint error by 2.33%, the mean acceleration difference by 15.01%, and the Fréchet gesture distance by 59.32%.

Index Terms: Computing methodologies—Virtual Agents; Computing methodologies—Co-Speech Gestures; Computer systems organization—Transformer

1 INTRODUCTION

People communicate face-to-face in the real world usually using gestures to complement the information. Thus, natural and co-speech gestures can make virtual agents more anthropomorphic. However, the automatic generation of co-speech gestures is a tricky problem due to the weak correlation between human gestures and other modalities (text, audio).

Researchers have proposed to generate co-speech gestures for speakers from multiple input modalities. Yoon et al. [5] proposed a multimodal gesture generation framework using audio, text, and speaker ids. Bhattacharya et al. [1] further used Mel Frequency Cepstrum Coefficient (MFCC) to extract audio features based on Yoon et al., which focused more on the representation of audio low-frequency information. However, these methods considered only a single audio modality (waveform or MFCC) and had a poor understanding of the semantics and beats of the audio. In addition, they extracted features by convolution, which leads to extracting only local features of the input modal. When switching to long audio or long text, they often produced gestures that do not match the general overall information. In [2], Transformer is used to generate gestures so that the generated gestures better match the semantics.

Inspired by the above discussion, we worked on generating co-speech upper-body gestures for virtual agents in VR environments. To model the weak correlation between the input modalities, we propose a Transformer-based model that encodes four-modal-like information. Different encoders are used to extract the features of different modal information. Mel Spectrogram encoder is based on Swin Transformer pre-trained model [4] to extract spectral features hierarchically. This enhances the understanding of the overall semantics and beats information of the audio. Audio waveform encoder is used to extract local features of audio waveform to enrich

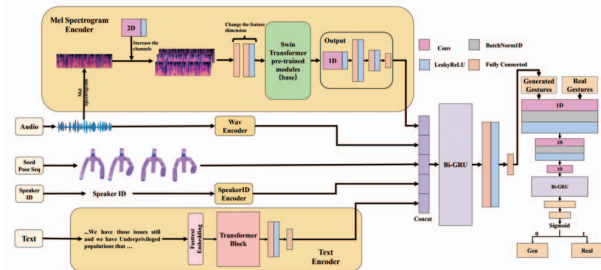


Figure 1: Overall architecture of the network model.

audio information. The Text encoder extracts text features based on a self-attention mechanism to further complement the semantic features of the audio. In addition, we encode SpeakerIDs to control the speaking style as Yoon et al. [5] to make the virtual agent more anthropomorphic. These allow the generated gestures to be synchronized with the audio, matching the semantic information, and producing natural results when switching between audio clips.

2 METHODS

Fig 1 shows the overall architecture of the network model. Our method can synthesize anthropomorphic gestures for virtual agents given the Audio waveforms, Text, SpeakerIDs. Then, we transform the waveform with Fourier transform and convert the spectrum to mel-scale to obtain Mel-Spectrogram. Finally, we get the four-modal-like information.

2.1 Encoding Multimodal Information

Waveform Encoder Given an audio waveform, We use a cascaded one-dimensional convolutional layer to extract features. Since the audio frequency is usually fixed, the audio waveform latent feature sequence with the same length as the output motion frame can be obtained by adjusting the step size and filling the size of the convolution layer.

Mel-Spectrogram Encoder We regard the Mel-Spectrogram as a kind of image with a number of channels of 1, height and width, respectively, time length and feature dimensions. And an encoder based on Swin-Transformer pre-trained model is designed to extract the features, which is further trained on the TED dataset [5]. We first preprocess this Mel-Spectrogram. Increase the number of channels by two-dimensional convolution, and increase the width and height dimensions by linear layers.

Text Encoder To keep the same length as the gestures, we first padded and embedded the subsequence following the method of Yoon et al. [5]. Next, we use a 2-layer transformer encoder followed by a fully connected layer to obtain potential features of the text modal.

SpeakerID Encoder For the speaker IDs, we use one-hot vectors. we use a set of fully connected layers to map it to the style embedding

*e-mail: yuyue@bit.edu.cn

†e-mail: 3120211058@bit.edu.cn

Table 1: Quantitative evaluation of results. Note that these three metrics are the lower the better. Bold indicates best.

Dataset	Method	MAJE(mm)	MAD(mm/s ²)	FGD
TED Gesture	GTC [5]	26.54	3.15	3.82
	S2AG [1]	24.49	2.93	3.54
	Ours	23.92	2.49	1.44

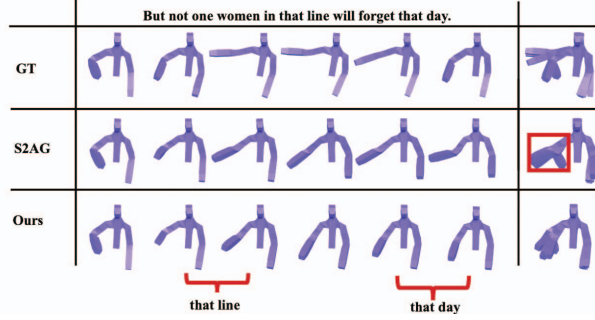


Figure 2: The qualitative experiment result compared with Ground Truth (GT) and S2AG.

space and capture the mean and log variance of the latent distribution of speaker styles.

2.2 Generator and Discriminator

Generator A multilayer Bi-GRU is used in the generator to solve the problem of asynchrony between audio and gestures. The four-like-modal encoded latent feature sequences and the seed gestures sequences are concatenated as the input to the Bi-GRU network, and we obtain the predicting gestures sequences.

Discriminator The discriminator is used to discriminate between generated and real gestures. For the discriminator, we also use a Bi-GRU, which takes in a gesture of sequence length T . We pass the gesture of sequence G through the Bi-GRU to obtain the discriminator embeddings sequence.

3 EXPERIMENTS AND RESULTS

3.1 Experiments

We trained our model in the TED Gesture dataset [5], which is a large-scale dataset. It contains different speakers, as well as the corresponding audio, text recordings, and human skeletal sequences. And large enough to learn the mapping between the input modalities and the human gestures. For generator and discriminator loss functions, we adopt the same approach as Yoon et al. [5].

3.2 Results

We compare our method with Gestures from Trimodal Context (GTC) [5] and Speech to Affective Gestures (SA2G) [1]. For evaluation metrics, We adopt evaluation metrics of mean absolute joint error (MAJE), mean acceleration (MAD), and the Fréchet Gesture Distance (FGD) [5] previously used in co-speech gesture generation for quantitative analysis. MAJE and MAD measure the predicted joint positions and joint accelerations for all time steps, joints, and samples against the ground truth mean and average ℓ^2 -norm errors. They denote the closeness of the predicted joint position to the ground truth joint position and the match between ground truth and predicted joint motion, respectively. The FGD is used to measure how close the generated gesture distribution is to the real gesture and is computed as the fréchet distance between the real gesture and the latent features of the generated gesture.

For a fair comparison, we used the feature extractors provided in [5]. We evaluated on the TED Gesture dataset. The results of the quantitative assessment are shown in Tab 1. We can see that MAJE, MAD, and FGD improved by 2.33%, 15.01%, and 59.32% respectively compared to the latest S2AG baseline. Among them, MAD and FGD have been significantly improved. Semantic expressions are usually displayed in terms of joint movements. The MAD metric is reduced, representing that the gestures generated by our method are more consistent with the semantic information corresponding to the audio. In addition, the FGD index has improved significantly, which indicates that our generated gesture distribution is very close to the real gesture distribution, and the synthetic gesture quality is high.

We also conducted qualitative experiments. First, we generated gestures on the TED-Gestures test dataset and sampled both methods at the same frame rate. The results are shown in Fig 2. According to the results, both methods generate gestures that match the semantics of speech more closely. For example, "... not one woman in that line ...", since the voice is more emotional, the gestures are larger, and the right arm points to match the semantics of "that day". However, the S2AG has a tremor at the elbow joint that leads to unnatural transition gestures. Our approach has a natural transition between gestures and matches the audio beat. Secondly, we also compared the generation of the two methods on the TSG dataset [3]. For the generation results, please see the supplemental video.

4 CONCLUSION

In this paper, we propose a novel framework for co-speech gesture generation for virtual agents from four-like-modalities based on Transformer. A Swin Transformer pre-trained model is used to extract audio spectral features hierarchically. It is also combined with audio waveform features as audio representations to enrich audio local features. In addition, we use a self-attentive mechanism to encode text modal so that the generated gestures matches the overall semantic content of the speech text. Co-speech gestures synthesized by our framework achieve the best results on multiple evaluation metrics. We also have the limitation that we did not use a 3D dataset to train our model, but recovered skeletal motion sequences from videos as training data, which are richer and more diverse in style. In the future, we can consider 3D datasets with multiple styles to further train our model.

ACKNOWLEDGMENTS

This work is supported by National Natural Science Foundation of China (61807002).

REFERENCES

- [1] U. Bhattacharya, E. Childs, N. Rewkowski, and D. Manocha. Speech2affectivegestures: Synthesizing co-speech gestures with generative adversarial affective expression learning. 2021.
- [2] U. Bhattacharya, N. Rewkowski, A. Banerjee, P. Guhan, A. Bera, and D. Manocha. Text2gestures: A transformer-based network for generating emotive body gestures for virtual agents. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pp. 1–10. IEEE, 2021.
- [3] T. Kucherenko, P. Jonell, Y. Yoon, P. Wolfert, and G. E. Henter. A large, crowdsourced evaluation of gesture generation systems on common data: The genea challenge 2020. 2021.
- [4] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 10012–10022, 2021.
- [5] Y. Yoon, B. Cha, J.-H. Lee, M. Jang, J. Lee, J. Kim, and G. Lee. Speech gesture generation from the trimodal context of text, audio, and speaker identity. *ACM Trans. Graph.*, 39(6), nov 2020. doi: 10.1145/3414685.3417838